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$$\int \frac{dx}{4x^2 + 1}$$
$$= \int \frac{dx}{(2x)^2 + 1}$$

$$\text{Stel } u = 2x \Rightarrow du = 2dx \Rightarrow \frac{1}{2}du = dx$$

$$= \frac{1}{2} \int \frac{du}{u^2 + 1}$$

$$= \frac{1}{2} \text{Bgtan}(u) + C = \frac{1}{2} \text{Bgtan}(2x) + C$$

References